

LEADED SOLDER TYPES

Comprehensive Guide for SnPb Alloys in PCBA Manufacturing

Sn63Pb37 | Sn60Pb40 | Sn62Pb36Ag2 | Eutectic & Non-Eutectic Alloys

■■ RoHS RESTRICTED — For Exempt Applications Only

Medical | Military | Aerospace | High-Reliability Legacy | Repair of Pre-RoHS Products

1. INTRODUCTION TO LEADED SOLDER

Leaded solder, primarily tin-lead (SnPb) alloys, was the industry standard for electronics assembly for over 50 years before RoHS restrictions. Despite environmental concerns, it remains critical in exempt applications where reliability and proven performance are paramount. This guide covers all major SnPb alloy types, their properties, and process parameters.

Why Leaded Solder Still Matters:

- **Lower melting point (183°C eutectic):** Reduces thermal stress on components and PCBs
- **Superior wetting:** Spreads and flows better on copper surfaces than lead-free alloys
- **Shiny joint appearance:** Easier visual inspection and cosmetic acceptance
- **Proven long-term reliability:** Decades of field data in harsh environments
- **Lower process cost:** No special equipment modifications needed

■■ **Regulatory Status:** SnPb solder is restricted under RoHS Directive 2011/65/EU for most consumer electronics. It may only be used in exempt categories: medical devices, monitoring and control instruments, military/aerospace, and repair of equipment placed on the market before July 2006.

2. MAJOR LEADED SOLDER ALLOYS

2.1 Sn63Pb37 — Eutectic Tin-Lead

Sn63Pb37 is the most common eutectic leaded solder alloy. The 63% tin / 37% lead composition provides a single, sharp melting point at **183°C** with no plastic range. This makes it ideal for all soldering processes where minimal thermal exposure is desired.

Properties:

- **Melting Point:** 183°C (eutectic — liquidus = solidus)
- **Typical Reflow Peak:** 210–230°C
- **Wave Solder Pot:** 240–250°C
- **Wetting Speed:** Excellent — fastest of all common alloys
- **Joint Appearance:** Bright, shiny, smooth surface
- **Shear Strength:** ~25 MPa (adequate for most applications)
- **Thermal Fatigue:** Good, but inferior to SAC305 in high-cycling applications

Applications:

- General-purpose SMT and through-hole assembly
- Medical devices (exempt from RoHS)
- Military and aerospace electronics (MIL-STD compliance)
- Repair and rework of legacy equipment
- Prototype and low-volume production

■ **Process Advantage:** The low melting point allows faster conveyor speeds in wave soldering and shorter reflow profiles, increasing throughput while reducing energy consumption.

2.2 Sn60Pb40 — General Purpose

Sn60Pb40 is a near-eutectic alloy with a slightly wider melting range (183–190°C). The extra lead content makes it slightly softer and more ductile than Sn63Pb37. It was historically the most economical choice when lead was cheaper than tin.

Properties:

- **Melting Range:** 183–190°C (non-eutectic, small plastic range)
- **Wetting:** Very good — nearly equal to Sn63Pb37
- **Joint Appearance:** Bright and shiny
- **Ductility:** Higher than Sn63Pb37 (more lead = softer alloy)
- **Cost:** Slightly lower than Sn63Pb37 historically

Applications:

- General plumbing and electrical soldering (non-PCBA)
- Cable and wire harness assembly
- Sheet metal and stained glass work
- Lower-cost consumer electronics (pre-RoHS legacy)

2.3 Sn62Pb36Ag2 — Silver-Bearing Leaded

Sn62Pb36Ag2 (often called "SN62" or "leaded silver solder") contains 2% silver to prevent **silver scavenging** when soldering to silver-plated components or PCBs. Without silver, SnPb alloys dissolve silver pads and traces, weakening the joint.

Properties:

- **Melting Point:** 179°C (slightly lower than Sn63Pb37)
- **Wetting:** Excellent — silver improves fluidity
- **Silver Scavenging:** Eliminated — safe for Ag-plated surfaces
- **Mechanical Strength:** Higher than Sn63Pb37 due to Ag content
- **Cost:** Higher than standard SnPb due to silver content

Applications:

- Soldering to silver-plated PCBs or component leads
- High-reliability military and aerospace assemblies
- RF and microwave circuits with silver traces
- Switch and relay contacts with silver plating

■■ **Important:** Never use Sn63Pb37 on silver-plated surfaces. The lead dissolves silver rapidly, creating a brittle Sn-Ag intermetallic and leaving a weak, porous joint. Always use Sn62Pb36Ag2 for silver-bearing substrates.

2.4 Other Lead-based Alloys

Sn50Pb50

Higher lead content (50%) creates a wider melting range (183–212°C) and a very soft, ductile alloy. Used for applications requiring maximum joint flexibility or where lower tin content is desired to reduce tin pest (allotropic transformation of tin at low temperatures).

Sn40Pb60 & Sn30Pb70

Very high lead content alloys with melting points above 183°C (up to 275°C for Sn30Pb70). These are used for **step-soldering** — soldering components at progressively lower temperatures so previously soldered joints don't remelt. Also used for plumbing and non-electrical applications.

Sn63Pb37 + Dopants

Some specialty SnPb solders include antimony (Sb) or bismuth (Bi) dopants to improve mechanical properties or reduce melting point. Sn63Pb37Sb, for example, increases creep resistance for high-temperature applications.

3. SOLDER FORMS

3.1 Solder Paste (Sn63Pb37)

Leaded solder paste uses Type III (25–45µm) or Type IV (20–38µm) powder in a flux medium. It offers excellent printability, long stencil life, and wide process windows. No-clean, water-soluble, and RMA (rosin mildly activated) flux types are available.

- **Storage:** 0–10°C; 6-month shelf life
- **Stencil Life:** 8–12 hours at room temperature
- **Reflow Peak:** 210–230°C (well below SAC305)

3.2 Solder Bar (Wave & Dip)

Solid Sn63Pb37 or Sn60Pb40 bars loaded into wave solder pots. Operating temperature 240–250°C. Lower pot temperatures extend equipment life and reduce dross formation compared to lead-free.

- **Dross Rate:** ~50% less than SAC305 at equivalent throughput
- **Copper Dissolution:** Low and predictable
- **Pot Maintenance:** Weekly analysis; less frequent than lead-free

3.3 Solder Wire (Hand Soldering)

Flux-cored wire in diameters 0.3mm to 1.6mm. Rosin-core (R, RMA, RA) is standard for electronics. Acid-core is for plumbing ONLY and must never be used on PCBs. SAC305 wire is NOT a substitute for SnPb in exempt applications.

- **Tip Temperature:** 315–350°C (lower than lead-free)
- **Dwell Time:** 2–5 seconds (shorter than lead-free)
- **Tip Life:** 2–3× longer than lead-free soldering

3.4 Solder Preforms

Precision shapes (rings, washers, squares) for die-attach, connector soldering, and thermal management. Sn63Pb37 preforms are common in military hybrids and power module assembly.

4. ALLOY COMPARISON TABLE

Alloy	Composition	Melt Point	Cost	Best Use
Sn63Pb37	63% Sn, 37% Pb	183°C (Eutectic)	1.0×	General SMT/PTH
Sn60Pb40	60% Sn, 40% Pb	183–190°C	0.95×	General purpose / Plumbing
Sn62Pb36Ag2	62% Sn, 36% Pb, 2% Ag	179°C	1.4×	Silver-plated surfaces
Sn50Pb50	50% Sn, 50% Pb	183–212°C	0.85×	Ductile joints / Step solder
Sn40Pb60	40% Sn, 60% Pb	183–234°C	0.75×	Step soldering (2nd pass)
Sn30Pb70	30% Sn, 70% Pb	183–275°C	0.65×	High-temp / Plumbing
SAC305	96.5% Sn, 3% Ag, 0.5% Cu	217°C	3.0×	RoHS Compliant

Selection Guide: Use Sn63Pb37 for standard exempt electronics. Use Sn62Pb36Ag2 for silver-plated components. Use Sn50Pb50 or lower tin content for step-soldering sequences or applications requiring maximum ductility.

5. PROCESS GUIDELINES

5.1 Reflow Profile (Sn63Pb37 Paste)

Leaded reflow is simpler and more forgiving than lead-free due to the lower melting point:

Zone	Temperature	Time / Rate
Preheat	100–150°C	60–90 sec
Soak	150–183°C	60–90 sec
Reflow Peak	210–230°C	30–60 sec above 183°C
Cooling	< 6°C/sec	Down to 100°C

5.2 Wave Soldering

- **Pot Temperature:** 240–250°C
- **Contact Time:** 2–4 seconds
- **Conveyor Angle:** 5–7°
- **Flux:** Any standard electronics flux (RMA, no-clean, water-washable)
- **Preheat:** 80–120°C top side

5.3 Hand Soldering

- **Tip Temperature:** 315–350°C
- **Dwell Time:** 2–5 seconds
- **Tip Life:** Significantly longer than lead-free
- **Wire:** Sn63Pb37 or Sn62Pb36Ag2 with rosin flux core
- **Technique:** Standard touch-and-feed method

Energy Savings: Leaded soldering consumes approximately **30–40% less energy** than lead-free processes due to lower peak temperatures and shorter heating times.

6. QUALITY & INSPECTION

6.1 Visual Inspection (IPC-A-610)

Leaded joints have a **bright, shiny, smooth** appearance that is easy to inspect visually. IPC-A-610 acceptance criteria:

- **Acceptable:** Bright, smooth fillet with concave meniscus
- **Acceptable:** Slight graininess on large joints (normal)
- **Reject:** Dull, grainy, or frosted appearance (overheating or contamination)
- **Reject:** Cracks, voids, or insufficient wetting

6.2 Common Defects

- **Cold Joint:** Dull, rough surface — insufficient heat or movement during cooling
- **Bridging:** Excessive solder between adjacent pins/pads
- **Insufficient Solder:** Fillet height below 25% of lead thickness
- **Lifted Pad:** PCB pad delamination (check PCB quality, not solder issue)
- **Tin Pest:** Rare allotropic transformation at $< 13^{\circ}\text{C}$ in high-tin alloys

6.3 Intermetallic Compound (IMC)

SnPb forms Cu_6Sn_5 and Cu_3Sn intermetallic layers at the solder-copper interface. Target IMC thickness: $1\text{--}3\mu\text{m}$. Excessive IMC ($> 5\mu\text{m}$) from overheating creates a brittle joint prone to fracture. Leaded solders generally form thinner, more stable IMC layers than SAC305.

7. SAFETY & ENVIRONMENTAL HANDLING

7.1 Health Hazards

Lead is a neurotoxin. Proper handling procedures are mandatory:

- **Wash hands** thoroughly after handling solder
- **Do NOT eat, drink, or smoke** in soldering work areas
- **Use fume extraction** — solder fumes contain lead oxide and flux vapors
- **Wear gloves** when handling bare solder or dross
- **Eye protection** required during wave pot skimming and bar loading

7.2 Waste Disposal

- **Dross:** Collect in labeled hazardous waste containers
- **Spent pots:** Recycled through certified metal reclamation facilities
- **Wipes and rags:** Disposed as lead-contaminated waste
- **Never dispose** of leaded solder in general trash or drains

7.3 Workplace Separation

If both leaded and lead-free lines exist in the same facility, implement strict separation:

- **Physical separation:** Different rooms or clearly marked zones
- **Color coding:** RED for leaded, GREEN for lead-free
- **Dedicated tools:** Irons, pots, stencils, and squeegees must not mix
- **Labeling:** All solder bars, paste jars, and wire spools clearly marked
- **Training:** All operators trained on cross-contamination risks

■■ **Cross-Contamination Risk:** Mixing even small amounts of SnPb into a lead-free pot voids RoHS compliance and can cause catastrophic joint failures in SnBi alloys. Always verify alloy type before loading any solder into equipment.

8. QUICK REFERENCE

For Standard Exempt Electronics:

→ **Sn63Pb37** — Eutectic, easiest to use, best wetting

For Silver-Plated Components:

→ **Sn62Pb36Ag2** — Prevents silver scavenging, slightly lower melt point

For Step-Soldering:

→ **Sn50Pb50** (1st pass) then **Sn63Pb37** (2nd pass) or lower tin alloys

For Plumbing / Non-Electrical:

→ **Sn50Pb50** or **Sn40Pb60** — wider melting range, more ductile

For Maximum Tip Life:

→ Lead soldering extends iron tip life 2–3× vs SAC305

References:

- IPC-A-610: Acceptability of Electronic Assemblies
- J-STD-001: Requirements for Soldered Electrical and Electronic Assemblies
- RoHS Directive 2011/65/EU Exemptions List
- Kester Solder Technical Handbook
- MIL-STD-202 & MIL-STD-883 (Military Testing Standards)

End of Document — Lead Solder Types Guide for PCBA Manufacturing